Student Academic Performance Analysis and At-Risk Student Identification

2026-11-03

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Abstract

Student academic performance is an important indicator of educational outcomes. Identifying students who may be at risk of poor academic performance can help educational institutions develop appropriate interventions and allocate academic support more effectively.

This study examines student academic performance using the Mathematics student performance dataset obtained from the UCI Machine Learning Repository. The dataset contains 395 student observations and 33 explanatory and outcome variables covering demographic, family, educational, social and lifestyle characteristics.

The analysis applies descriptive statistics, exploratory data analysis, hypothesis testing, multiple linear regression and logistic regression. Students scoring below 10 in the final examination grade were classified as academically at risk.

The analysis identified previous academic failures, age, school support and frequency of going out as statistically significant predictors in the weighted logistic regression model. The weighted model achieved a test-set sensitivity of 61.54%, specificity of 77.36%, balanced accuracy of 69.45% and an area under the ROC curve of 0.7562.

The findings demonstrate the potential of statistical modelling for investigating student performance and developing early-warning approaches. However, the results should be interpreted cautiously because of the relatively small test sample, observational nature of the dataset and limitations in predictive performance.

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

Academic performance is influenced by several interacting factors, including previous academic achievement, student characteristics, family background, school environment, social activities and study habits.

Educational institutions require reliable approaches for understanding these factors and identifying students who may experience academic difficulties. Statistical analysis provides methods for investigating relationships between academic outcomes and explanatory variables.

Regression modelling is particularly useful because it allows researchers to examine how several explanatory variables are associated with an outcome while accounting for other variables included in the model.

Predictive modelling can also support early identification of students who may benefit from additional academic assistance.

This study applies statistical techniques to investigate student performance and develop a logistic regression model for identifying students classified as academically at risk.

1.2 Problem Statement

Poor academic performance can affect students' educational progression and future opportunities. However, identifying students who may experience academic difficulties requires systematic examination of relevant academic, demographic and social characteristics.

Without appropriate analytical approaches, educational institutions may find it difficult to identify patterns associated with poor academic outcomes.

This study addresses this problem by applying statistical analysis and predictive modelling to student performance data. It investigates the relationships between selected student characteristics and final examination grades and evaluates the ability of logistic regression to classify students according to academic risk.

1.3 General Objective

To analyse student academic performance and develop a statistical model for identifying students who may be at risk of poor academic performance.

1.4 Specific Objectives

1. To examine the distribution and characteristics of student academic performance.
2. To investigate associations between student characteristics and final academic grades.
3. To identify factors associated with academic risk.
4. To develop a multiple linear regression model for explaining variation in final examination grades.
5. To develop and evaluate logistic regression models for identifying academically at-risk students.

1.5 Research Questions

1. What is the distribution of students' final examination grades?
2. Are selected demographic, educational and social characteristics associated with final academic performance?
3. Which factors are significantly associated with academic risk?
4. How well can logistic regression distinguish students classified as at risk from those not classified as at risk?

1.6 Significance of the Study

The study demonstrates how statistical techniques can be used to understand academic performance and identify patterns associated with academic risk.

The findings may be useful to:

- Educational institutions interested in monitoring academic performance.
- Researchers investigating educational outcomes.
- Academic advisers developing student-support strategies.
- Data analysts applying statistical modelling to educational datasets.

The project also demonstrates practical application of R programming, statistical inference, regression modelling and predictive analytics.

CHAPTER TWO: LITERATURE AND CONCEPTUAL BACKGROUND

2.1 Academic Performance

Academic performance refers to the extent to which students achieve expected educational outcomes. Examination grades are commonly used as measurable indicators of academic achievement.

However, academic performance may be influenced by several factors. These include previous academic achievement, study habits, family circumstances, school-related support and student behaviour.

2.2 Statistical Modelling in Education

Statistical models provide a systematic framework for examining relationships between student characteristics and academic outcomes.

Linear regression is suitable when the outcome is numerical, such as a final examination grade.

Logistic regression is appropriate when the outcome is categorical, such as whether a student is classified as academically at risk.

2.3 Early Identification of Academic Risk

An early-warning approach attempts to identify students who may require additional academic support.

Such a system should be treated as a decision-support tool rather than a mechanism for labelling students permanently. Predictions should be interpreted alongside individual circumstances and professional educational judgement.

CHAPTER THREE: DATA AND METHODOLOGY

3.1 Data Source

The study uses the Mathematics student performance dataset from the UCI Machine Learning Repository.

The dataset is available at:

https://archive.ics.uci.edu/dataset/320/student+performance

The dataset contains information on student demographics, family background, school characteristics, social activities, study habits, absences and academic grades.

3.2 Dataset Description

The Mathematics dataset contains 395 observations and 33 variables.

The principal outcome variable is G3, representing the final examination grade on a scale from 0 to 20.

Other academic variables include:

- G1: First-period grade.
- G2: Second-period grade.
- G3: Final-period grade.

Additional variables describe student characteristics, family circumstances, school support and lifestyle factors.

3.3 Data Preparation

The dataset was imported into R using the `read.csv()` function with a semicolon delimiter.

```
## [1] 395  34
```

Table 1: Dataset Structure and Data Quality

Observations	Variables	Missing_Values	Duplicate_Rows
395	34	0	0

The dataset contained 395 observations and 33 variables. Initial inspection identified no missing values and no duplicated rows.

3.4 Definition of Academic Risk

Students were classified according to their final examination grade.

A student with G3 below 10 was classified as At Risk, while a student scoring 10 or above was classified as Not At Risk.

This threshold was adopted as an operational classification rule for the study. It should not be interpreted as a formal diagnosis of academic difficulty.

Table 2: Distribution of Academic Risk Categories

risk_status	n	Percentage
At Risk	130	32.91
Not At Risk	265	67.09

The classification resulted in:

- 130 students classified as At Risk (32.91%).
- 265 students classified as Not At Risk (67.09%).

3.5 Statistical Methods

The following methods were applied:

1. Descriptive statistics to summarise the dataset.
2. Exploratory data analysis to investigate patterns and relationships.
3. Hypothesis testing to examine selected associations.
4. Multiple linear regression to investigate variation in final grades.
5. Logistic regression to classify academic risk.
6. Confusion matrices and ROC analysis to evaluate predictive performance.

A significance level of 5% was adopted for statistical hypothesis testing.

CHAPTER FOUR: EXPLORATORY DATA ANALYSIS

4.1 Descriptive Statistics of Final Grades

The final grade variable G3 was examined using measures of central tendency and dispersion.

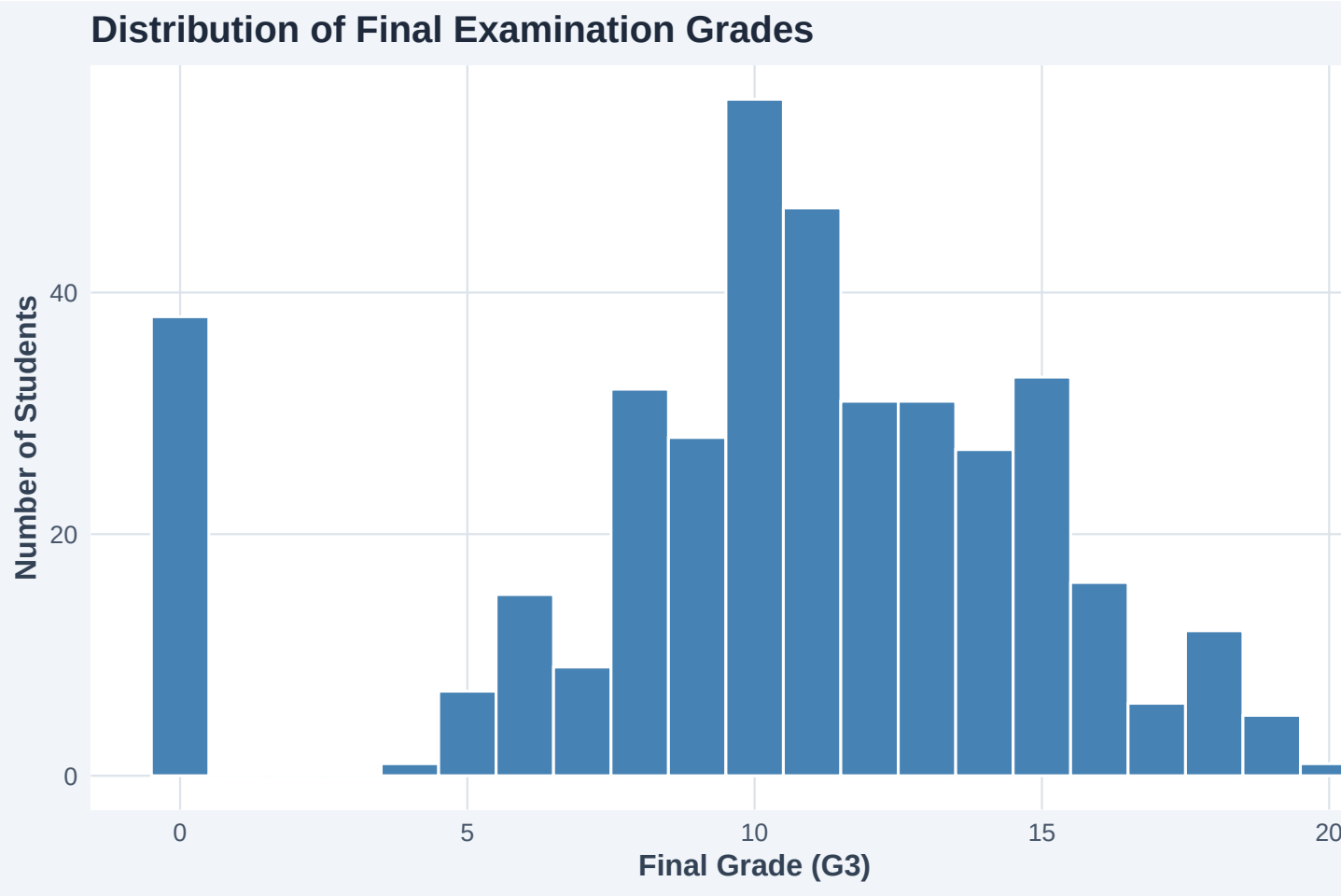
Table 3: Descriptive Statistics of Final Grades

Mean	Median	Standard_Deviation	Minimum	Maximum	Skewness
10.415	11	4.581	0	20	-0.727

The final grade had a mean of approximately 10.42 and a median of 11. The standard deviation was approximately 4.58, with grades ranging from 0 to 20.

The distribution had negative skewness of approximately -0.73, indicating some concentration of observations towards the higher grade values, with a tail extending towards lower grades.

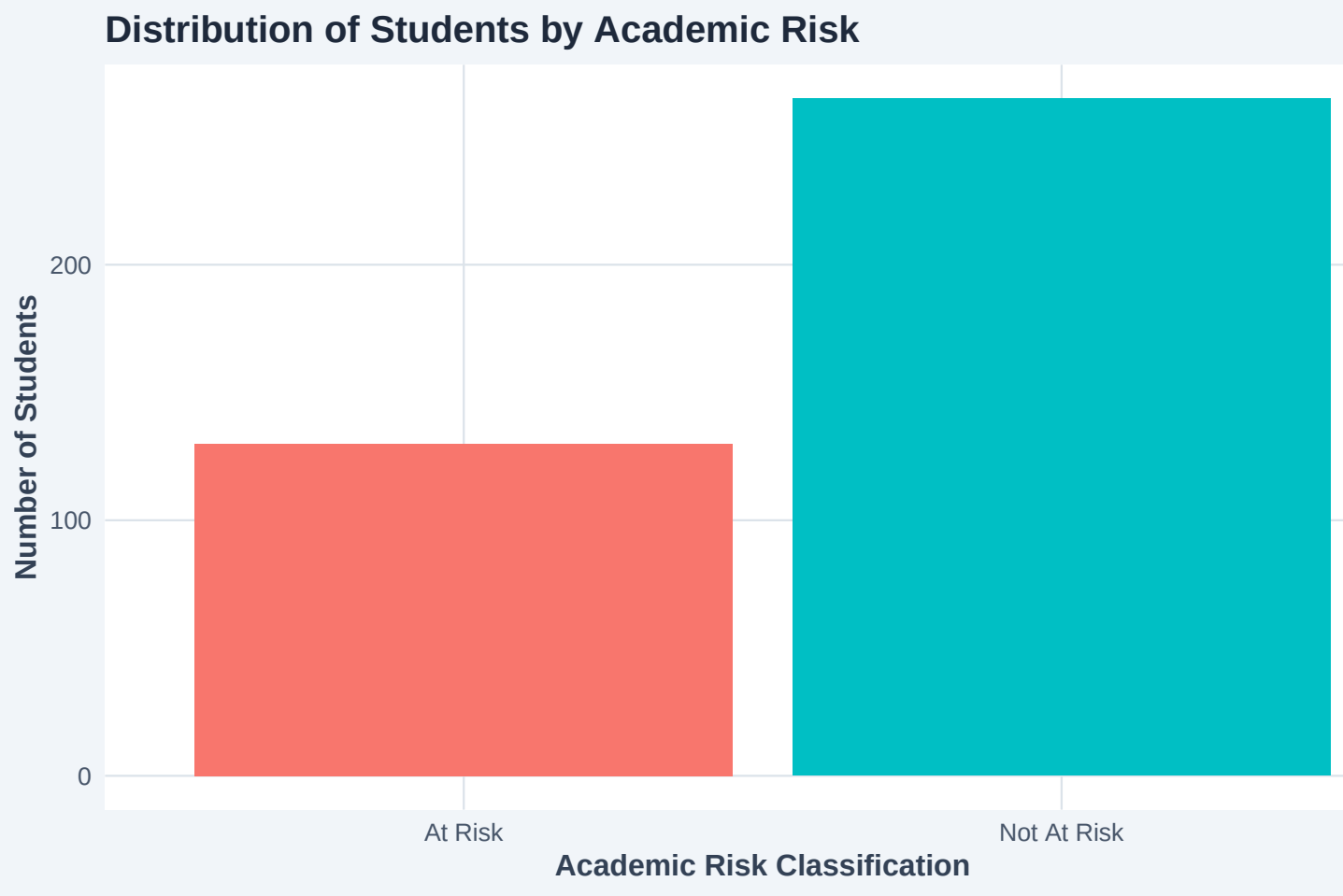
4.2 Distribution of Final Grades



The histogram provides an overview of student achievement and reveals the distribution of final examination grades.

The presence of low and zero grades highlights the importance of examining students who may require additional academic support.

4.3 Academic Risk Distribution

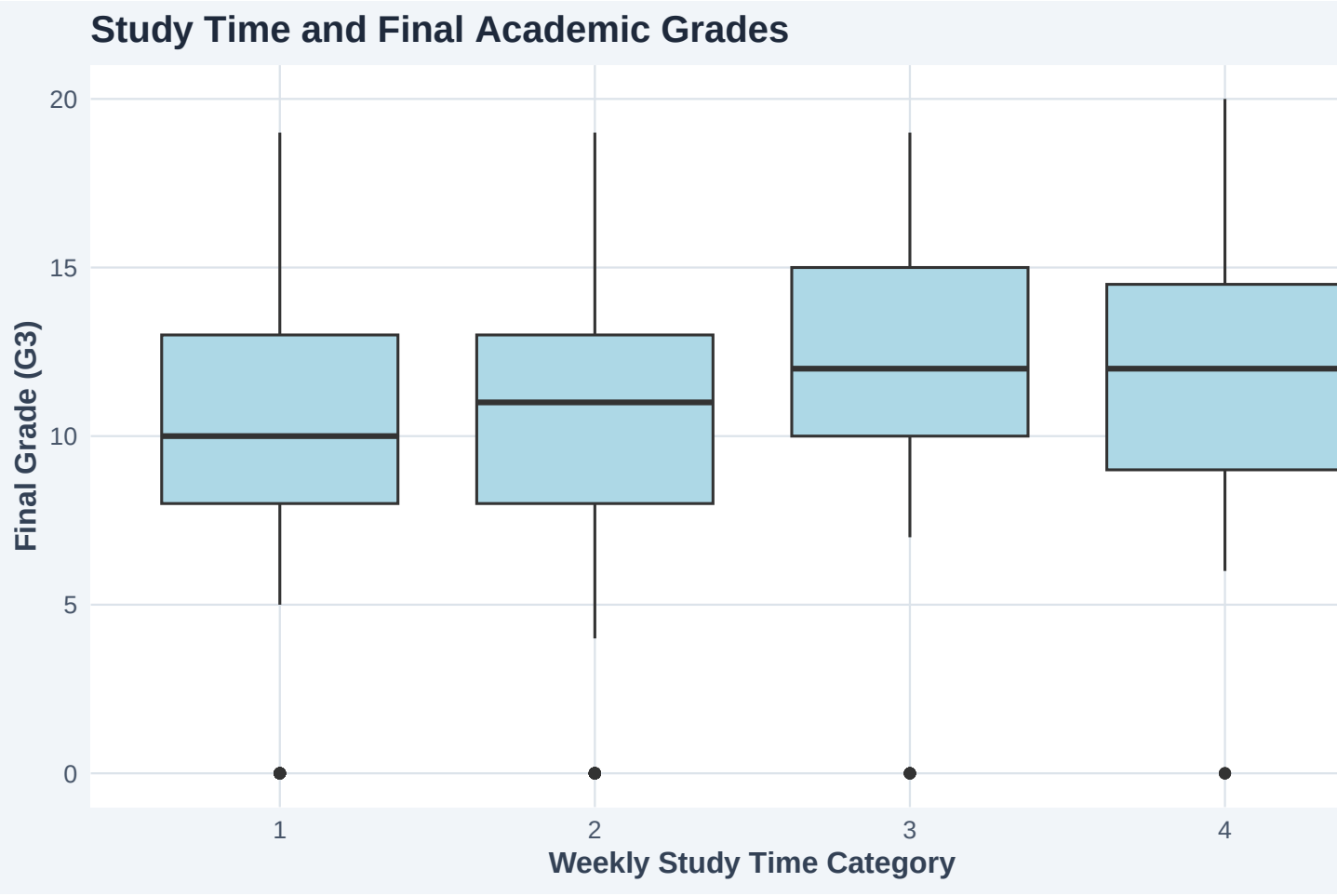


Approximately one-third of the students were classified as academically at risk under the study's chosen threshold.

This represents a substantial proportion of the sample and motivates further investigation into the characteristics associated with academic risk.

4.4 Study Time and Academic Performance

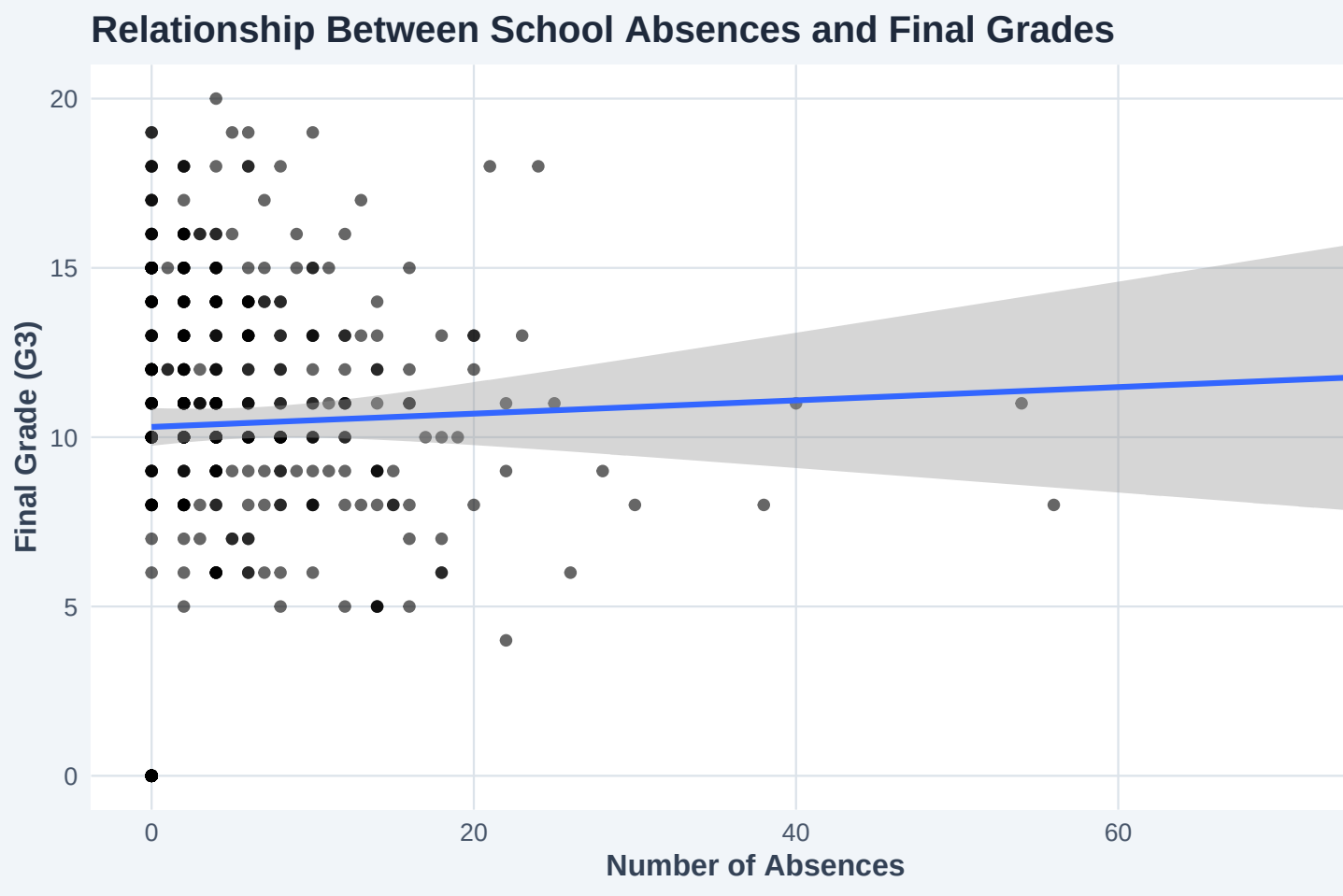
Study time was examined in relation to final examination grades.



The Spearman rank correlation between study time and final grade was approximately 0.105, with a p-value of 0.0367.

Although statistically significant at the 5% level, this represents a very weak positive association. Therefore, the result does not indicate a strong relationship between the recorded study-time category and final academic performance.

4.5 Absences and Academic Performance



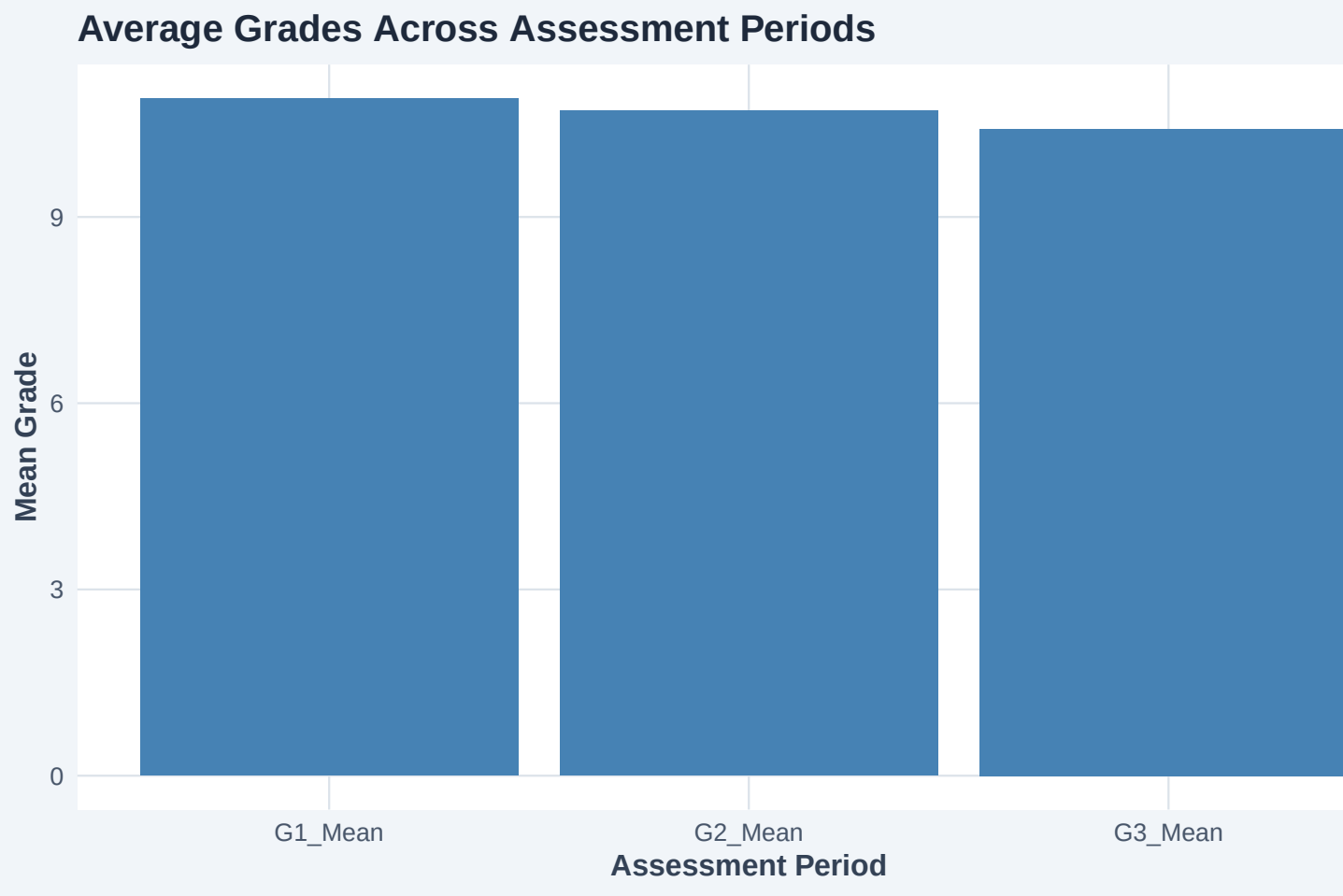
The Pearson correlation between absences and final grades was approximately 0.034, with a p-value of 0.497.

The Spearman correlation was approximately 0.0177, with a p-value of 0.725.

Both results indicate that there was insufficient statistical evidence of a meaningful association between absences and final grades in this sample.

The absence of a statistically significant correlation does not establish that attendance is unimportant in other educational settings.

4.6 Previous Academic Grades



Previous academic grades were investigated because they may provide information about students' academic progression.

The subsequent regression analysis examines the contribution of G1 and G2 to final grade variation.

CHAPTER FIVE: STATISTICAL HYPOTHESIS TESTING

5.1 Study Time and Final Grade

The Spearman rank correlation was used to investigate the association between study time and final grade.

The hypotheses were:

H0: There is no monotonic association between study time and final grade.

H1: There is a monotonic association between study time and final grade.

The observed Spearman correlation was approximately 0.105, with $p = 0.0367$.

Since the p-value was below 0.05, the null hypothesis was rejected. However, the association was very weak in magnitude.

5.2 Absences and Final Grade

The hypotheses were:

H0: There is no linear association between absences and final grade.

H1: There is a linear association between absences and final grade.

The Pearson correlation was approximately 0.034, with p = 0.497.

Since the p-value exceeded 0.05, the null hypothesis was not rejected. The data did not provide sufficient evidence of a linear association between absences and final grade.

5.3 Comparison of Earlier Grades Between Risk Groups

The Wilcoxon rank-sum test was used to compare G1 and G2 between students classified as At Risk and Not At Risk.

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  G1 by risk_status
## W = 2645.5, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  G2 by risk_status
## W = 942.5, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
```

The earlier grade distributions differed significantly between the two academic-risk groups.

This finding is consistent with the importance of previous academic achievement in predicting subsequent academic outcomes.

CHAPTER SIX: MULTIPLE LINEAR REGRESSION

6.1 Model Specification

Multiple linear regression was used to examine factors associated with final examination grades.

The model was specified as:

G3 = 0 + 1G1 + 2G2 + 3Studytime + 4Failures + 5Absences + 6Age + 7Sex +

Where:

- G3 represents the final examination grade.
- G1 and G2 represent earlier examination grades.
- Studytime represents weekly study-time category.
- Failures represents previous class failures.
- Absences represents school absences.
- Age represents student age.
- Sex represents student sex.
- represents the error term.

6.2 Regression Results

```
##
## Call:
## lm(formula = G3 ~ G1 + G2 + studytime + failures + absences +
##     age + sex, data = student_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.0797 -0.4294  0.3140  0.9741  3.6787
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.02549    1.36829   0.749 0.454030
## G1           0.16237    0.05625   2.886 0.004117 **
## G2           0.96121    0.04955  19.399  < 2e-16 ***
## studytime   -0.13465    0.12353  -1.090 0.276355
## failures    -0.22669    0.14367  -1.578 0.115401
## absences     0.04164    0.01223   3.404 0.000733 ***
## age         -0.15859    0.07951  -1.994 0.046799 *
## sexM         0.16905    0.20478   0.826 0.409583
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.902 on 387 degrees of freedom
## Multiple R-squared:  0.8308, Adjusted R-squared:  0.8277
## F-statistic: 271.4 on 7 and 387 DF,  p-value: < 2.2e-16
```

The fitted model achieved:

- R-squared: 0.8308.
- Adjusted R-squared: 0.8277.
- Residual standard error: approximately 1.902.

The model therefore explained approximately 83.08% of the observed variation in final grades within this sample.

G1 had a positive coefficient of approximately 0.1624 and was statistically significant ($p = 0.0041$).

G2 had a positive coefficient of approximately 0.9612 and was highly statistically significant ($p < 0.001$).

Study time and previous failures were not statistically significant at the 5% level in this particular model.

Absences had a positive coefficient of approximately 0.0416 ($p = 0.000733$), while age had a negative coefficient of approximately -0.1586 ($p = 0.0468$).

These coefficients describe conditional associations in the fitted model and should not be interpreted as causal effects.

6.3 Multicollinearity Assessment

Variance inflation factors were examined to investigate multicollinearity among explanatory variables.

The VIF values for G1 and G2 were approximately 3.798 and 3.785, respectively. The remaining predictors had values close to 1.

These results do not indicate severe multicollinearity, although G1 and G2 share some expected correlation because both measure previous academic performance.

6.4 Regression Diagnostics

Regression diagnostics were conducted to investigate the assumptions of the linear regression model.

The diagnostic assessment indicated departures from normality of residuals and possible heteroscedasticity. Some observations also showed relatively high influence.

The Cook's distance screening identified 29 observations above the heuristic threshold of 4/n.

These observations were not automatically removed because influential observations may represent valid student records.

6.5 Sensitivity Analysis

A sensitivity analysis was conducted by refitting the model after excluding the 29 observations flagged by the Cook's distance screening.

The original model had an R-squared of approximately 0.8308, while the reduced-sample model had an R-squared of approximately 0.9464.

The improved fit after exclusion suggests that the flagged observations had substantial influence on the model fit.

However, the improvement does not justify automatically removing those records. The original full-sample model remains the primary analysis, while the reduced-sample model serves as a sensitivity comparison.

CHAPTER SEVEN: LOGISTIC REGRESSION AND ACADEMIC RISK

7.1 Logistic Regression Framework

Logistic regression was used to estimate the probability that a student belongs to the At Risk category.

The dependent variable was binary:

- 1: At Risk.
- 0: Not At Risk.

The logistic model estimates the log odds of academic risk as a function of selected student characteristics.

7.2 Avoiding Target Leakage

The earlier academic grades G1 and G2 were excluded from the early-warning logistic model.

This decision was made because the purpose of the model was to investigate whether demographic, family, educational and lifestyle characteristics could help identify risk without directly using earlier examination grades.

The exclusion helps avoid relying on information that is very close to the final academic outcome.

7.3 Full Logistic Regression Model

The full logistic regression model initially included a broad set of available student characteristics.

The analysis identified age, previous failures, school support, family support and frequency of going out as statistically significant in the full model.

The results suggested that these characteristics were associated with the odds of academic risk, conditional on the other included variables.

Support-related variables require particularly careful interpretation. Students receiving school or family support may already have been identified as experiencing academic difficulties. Therefore, the observed associations do not mean that support causes academic risk.

7.4 Reduced Logistic Regression Model

A reduced logistic regression model was fitted using selected predictors.

The model included:

- Age.
- Previous failures.
- Study time.
- School support.
- Family support.
- Frequency of going out.
- Absences.
- Desire for higher education.

The fitted model produced the following important results:

Predictor	Odds ratio	p-value
Age	1.291	0.0146
Previous failures	2.327	0.00000227
Study time	0.935	0.651
School support	2.516	0.00783
Family support	1.597	0.065
Going out	1.356	0.00457
Absences	1.010	0.476
Higher education aspiration	0.505	0.225

At the 5% significance level, age, previous failures, school support and going out were statistically significant.

Previous failures had an odds ratio of approximately 2.327. This indicates that a one-unit increase in the recorded number of previous failures was associated with approximately 2.33 times the odds of being classified as at risk, holding the other model predictors constant.

School support had an odds ratio of approximately 2.516. This association may reflect the fact that support is often targeted towards students experiencing difficulties.

Family support was not statistically significant at the 5% level in this reduced model.

7.5 Weighted Logistic Regression

The dataset contained 130 at-risk students and 265 students classified as not at risk.

To give greater influence to the minority class during model fitting, inverse-frequency class weights were applied.

The weighted model was estimated using the training dataset.

The results indicated that age, previous failures, school support and going out remained statistically significant at the 5% level.

The weighting approach was intended to improve the identification of at-risk students rather than simply maximise overall accuracy.

CHAPTER EIGHT: MODEL EVALUATION

8.1 Training and Testing Strategy

The dataset was divided into training and testing subsets using an 80:20 stratified split.

A fixed random seed of 123 was used to support reproducibility.

The training dataset contained 316 observations, while the test dataset contained 79 observations.

The stratified split preserved the approximate class proportions in both subsets.

8.2 Confusion Matrix Comparison

The unweighted logistic regression model produced the following test-set classification results:

	Actual Not At Risk	Actual At Risk
Predicted Not At Risk	48	19
Predicted At Risk	5	7

The weighted model produced:

	Actual Not At Risk	Actual At Risk
Predicted Not At Risk	41	10
Predicted At Risk	12	16

The weighted model correctly identified 16 of the 26 at-risk students, compared with 7 identified by the unweighted model.

This represents a substantial improvement in identifying students classified as at risk, although the weighted model also generated more false positives.

8.3 Classification Performance

Metric	Unweighted model	Weighted model
Accuracy	69.62%	72.15%
Sensitivity	26.92%	61.54%
Specificity	90.57%	77.36%
Precision	58.33%	57.14%
Balanced accuracy	58.74%	69.45%
Kappa	0.203	0.382

The weighted model achieved better sensitivity and balanced accuracy.

Sensitivity is particularly important in an early-warning setting because it measures the proportion of genuinely at-risk students correctly identified by the model.

The improvement came with a reduction in specificity, meaning that more students who were not classified as at risk were incorrectly flagged.

8.4 ROC Curve and Area Under the Curve

ROC analysis was used to examine the models' ability to distinguish between students classified as at risk and those not classified as at risk across different probability thresholds.

The unweighted model achieved an AUC of approximately 0.7569.

The weighted model achieved an AUC of approximately 0.7562, with a 95% confidence interval of 0.6482 to 0.8641.

The two models therefore demonstrated very similar overall discrimination.

The weighted model's advantage was mainly its classification behaviour at the selected threshold, rather than a meaningful improvement in AUC.

8.5 Cross-Validation

Ten-fold cross-validation was conducted using the training data to examine model stability.

The earlier unweighted cross-validation analysis produced a mean ROC AUC of approximately 0.6964, with mean sensitivity of 0.8931 and mean specificity of 0.3727 at the default classification threshold.

The variation across folds indicates that model performance was not entirely stable.

A later weighted cross-validation run reported mean accuracy of approximately 0.6678 and Kappa of 0.2696. That run did not provide a cross-validated ROC estimate, so it should not be directly compared with the earlier cross-validated ROC result.

8.6 Interpretation of Predictive Performance

The weighted model appears more appropriate when the main objective is to identify a greater proportion of students who may require additional academic support.

However, its performance is not sufficiently strong to justify using predictions as the sole basis for educational decisions.

Predictions should instead guide further assessment and supportive intervention.

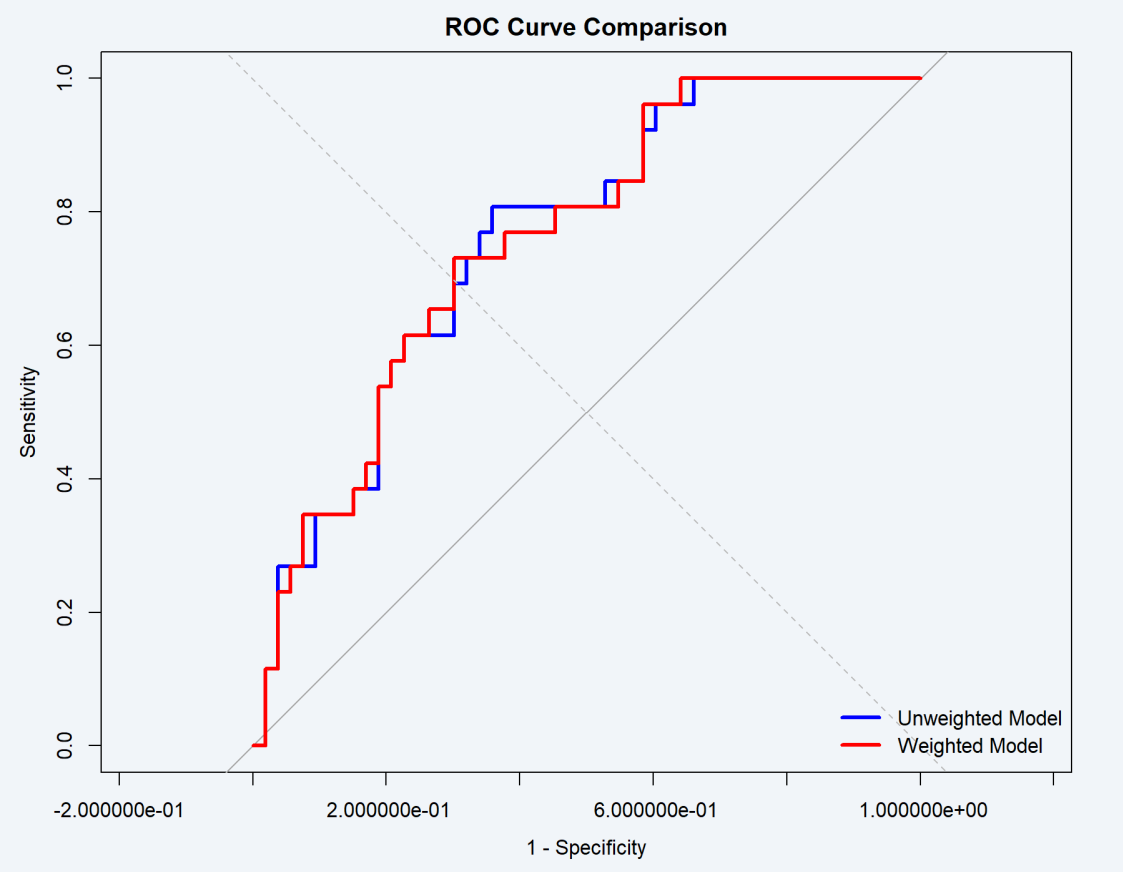


Figure 1: ROC curve comparison of unweighted and weighted logistic regression models

CHAPTER NINE: DISCUSSION, CONCLUSIONS AND RECOMMENDATIONS

Discussion of Findings

The findings demonstrate that student academic performance is associated with several academic, demographic and social characteristics.

The descriptive analysis showed that final examination grades varied considerably across students. Approximately 32.91% of the sample was classified as academically at risk under the selected threshold.

The correlation analysis identified a statistically significant but very weak positive association between study time and final grades. No statistically significant correlation was identified between absences and final grades.

The linear regression model explained approximately 83.08% of the variation in final grades. Earlier academic performance, particularly G2, contributed substantially to the model.

The logistic regression analysis identified previous failures as an important predictor of academic risk. Age, school support and going out were also statistically significant in the weighted model.

The relationship between school support and academic risk should not be interpreted as evidence that support negatively affects students. It may reflect targeted support provided to students who are already experiencing difficulties.

The weighted logistic regression model improved sensitivity from 26.92% to 61.54% at the 0.5 classification threshold. This demonstrates the value of considering class imbalance when developing models for early identification.

However, the weighted and unweighted models achieved nearly identical AUC values. This indicates that weighting changed the balance between sensitivity and specificity more than it changed overall ranking ability.

Study Limitations

1. **Dataset size:** The study used 395 observations, with only 79 observations in the test dataset. This limits the precision of predictive performance estimates.
2. **Observational data:** The analysis identifies associations but cannot establish causal relationships.
3. **Classification threshold:** The decision to classify grades below 10 as academic risk is an operational choice.
4. **Predictor limitations:** The dataset does not capture every factor that may influence academic performance.
5. **Model assumptions:** Diagnostic analysis identified potential departures from some linear regression assumptions.
6. **Predictive uncertainty:** The weighted model demonstrated moderate discrimination rather than perfect classification.
7. **Validation limitations:** The test set was used repeatedly during exploratory model and threshold comparisons. Consequently, the reported test performance should be regarded as exploratory rather than a fully independent final assessment.
8. **Generalisation:** Findings from this dataset should not automatically be generalised to all schools or educational systems.

Conclusions

This study applied statistical analysis and predictive modelling to investigate student academic performance and identify factors associated with academic risk.

The descriptive analysis showed that approximately one-third of the students were classified as at risk under the study's operational threshold.

The linear regression analysis demonstrated a strong relationship between earlier examination grades and final academic performance.

The logistic regression analysis identified previous failures, age, school support and frequency of going out as statistically significant predictors in the weighted model.

The weighted logistic regression model improved the identification of at-risk students compared with the unweighted model at the 0.5 classification threshold.

The findings demonstrate that statistical modelling can provide useful information for academic monitoring and early identification. Nevertheless, predictive results should be treated as decision-support information rather than definitive assessments of individual students.

Recommendations

Based on the findings, the study recommends:

1. **Academic monitoring:** Schools should monitor previous academic performance to identify students who may need additional support.
2. **Targeted interventions:** Students with repeated academic failures should receive appropriate academic guidance and follow-up.
3. **Support evaluation:** Schools should examine how academic support programmes are targeted and whether they effectively address student needs.
4. **Responsible predictive modelling:** Predictive models should support, rather than replace, professional judgement by teachers and academic advisers.
5. **Further research:** Future studies should use larger datasets and independent validation samples to improve the reliability and generalisability of predictive models.
6. **Model improvement:** Future work may investigate alternative algorithms, probability calibration and threshold selection using nested cross-validation.

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